



M/s HYUNDAI-WIA (INDIA) PVT. LTD.

Your Ref: RFQ: 1)180330_D-Shaft.
2) 180427_T-HSG & Outer Race.
And Visit and your discussion with our
Mr. Ko Jinseok

Our Ref :- Tulip:EIN112384
Outer Race:EIN112385
Barshaft:EIN112386

Date : 17/07/2018

Final Quotation for EFD make Induction Hardening Machine for T-HSG, O-Race & D-Shaft.

Dear Sir,

We thank you for your enquiry and we are pleased to submit our proposal for **Induction Hardening Machine for T-HSG, O-Race & D-Shaft.**

With reference to the visit made by our Mr. Ko Jinseok we are submitting our final prices for the respective Machine:-

Final price for each machines are:-

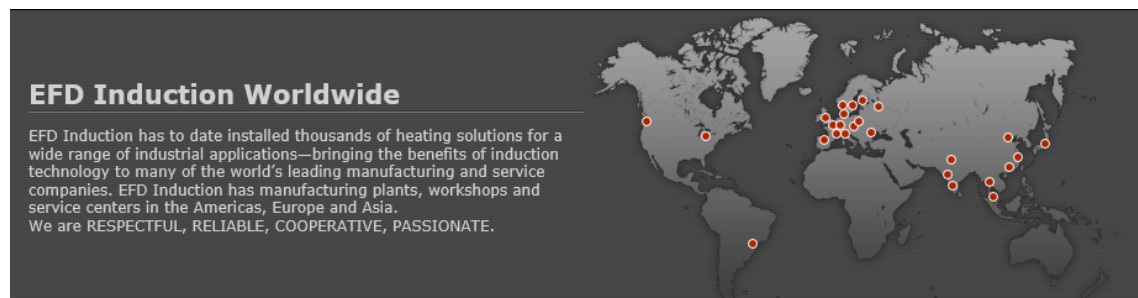
Sl No	Description	Quantity	Total Price	Quote Ref:-
1	D shaft- Induction Hardening M/c	1	INR 4,67,00,000/-	EIN112386
	Installation & Commission for D Shaft		INR 3,00,000/-	
2	T-HSG- Induction Hardening M/c	1	INR 4,99,40,000/-	EIN112384
	Installation & Commission for T-HSG		INR 3,60,000/-	
3	O Race- Induction Hardening M/c	1	INR 4,55,20,000/-	EIN112385
	Installation & Commission for O-Race		INR 1,80,000/-	
Total			INR 14,30,00,000/-	
(Rupees Fourteen Crore Thirty Lakhs Only)				

Please refer our respective Quotes for Technical and Commercial details below.

EFD is the European market leader in the field of induction surface heat treatment applications. Based on your requirement, we have proposed a package representing state-of-the-art technology in **Induction heating.**

EFD was founded in 1996 by the merger of two medium sized traditional companies (FDF and ELVA) and has approximately 1000 employees all over the world today.

EFD Induction Private Limited



Manufacturing plants are located in France, Norway, Germany, the USA, India, China and Romania. Furthermore, there are subsidiaries in Austria, Spain, the UK, Portugal, Sweden, Japan and Italy. We are represented in 51 countries.

EFD is the market leader in Europe in terms of delivery of installations for Induction Equipment and second worldwide.

More details can be found at <http://www.efd-induction.com>

EFD INDUCTION PRIVATE LIMITED, INDIA

EFD India (EIN) operations started in the year 1996. Today, EIN boasts of strength of more than 250 employees with a turnover of INR 840 Million. Since inception, EIN has installed and commissioned more than 750 numbers of Induction Heating Equipment in India and ASEAN, apart from our supplies to Argentina, Slovakia, Australia and Dubai.

EFD equipment in India are manufactured at Bangalore and supported with local branch sales and service offices in Pune and Gurgaon.

Applications catered to...

We specialize in the manufacture of solid-state power sources and equipment for various Induction Heating Applications such as Hardening, Annealing, Brazing, Soldering, Straightening, Shrink fitting etc.

We offer Induction heating package combining our expertise in power electronics, machine building and process know-how.

Quality control/quality assurance

For EFD, quality production means making the customer and his demands on our product the central interest of the company. At EFD, we give a strong impetus to the quality policies and standards.

EFD Induction Private Limited



Our Quality Management System for Induction Heating equipment at EIN is ISO 9001: 2008 compliant. We are also certified under ISO/ TS 16949: 2009 for Commercial Heat Treatment activities.

We take pride over our certification with OHSAS 18001: 2007 for Safety Management System & EN ISO 14001: 2004 for Environmental Management System.

Clientele:

EFD India has been partnering with major automotive companies offering solutions on continual basis.

While EFD Group's clientele includes the big names in the respective fields, EFD India's prestigious list of clients for Induction Hardening applications include

1. Drivelines Segment – GKN Drivelines- India, Mexico, Thailand, Malaysia and Australia, Nexteer Automotives, NTN- India, Thailand, NSK ABC, Hyundai- Wia , TOYOTA Kirloskar Auto Parts

2. Steering Segment – JTEKT (Formerly Sona Koyo Steering systems Ltd), ZF Steering systems.

3. Engine Segment – DMW-Patiala (Locomotive Camshafts) , Tata Motors, Ashok Leyland, Mahle Migma, Rane Engine Valves Ltd, Automotive Valves Ltd., Varroc , Volkswagen- Argentina, FAW- Volkswagen- Changchun, China, Musashi Auto, TOYOTA, Hero MotoCorp

4. Transmission Segment – Ashok Leyland, TVS Motor company Ltd , Sona Somic Lemforder, SFL – Hosur, SFL- Chennai, Amalgamation Valeo, Setco, EXEDY Clutches, Stumpp Scheule and Somappa, Clutch Auto Components, TIDC, LGB

5. Other Components – Parsan – Turkey, Bluelight Industries – Dubai, L & T Komatsu, Endurance Technologies, Tube Products of India, Bosch Nasik, Bosch Bangalore, Bosch Jaipur, Brakes India Limited, Bill Forge Sata Vikas and many more.

We hope our proposal will meet your techno-commercial expectations. Please feel free to check with us for any clarifications.

We thank you for your interest in EFD products and look forward to being of service to you.

Yours faithfully,
for EFD Induction Private Ltd.,

Suhas S Rao
Deputy Manager - Marketing & Customer Relations
EFD Induction Private Limited

M/s. D Shaft Induction Hardning Machine

INDUCTION HARDENING MACHINE FOR BAR SHAFTS.

Quotation Ref: EIN112386

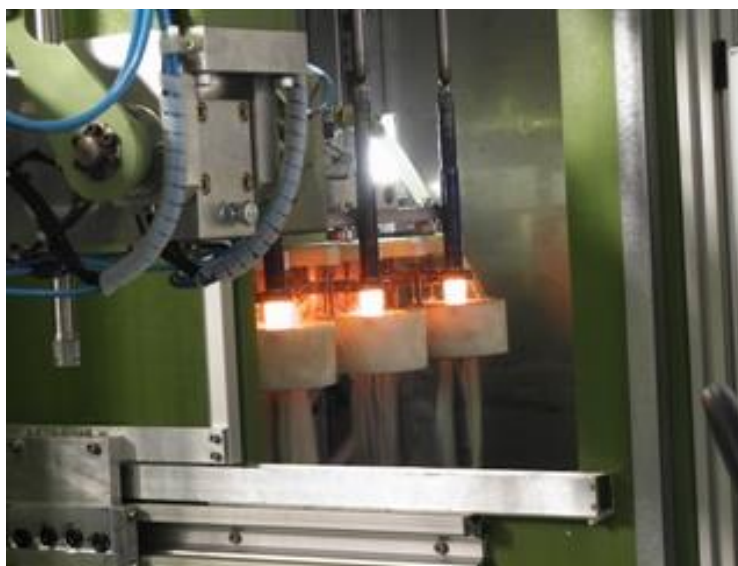


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ENQUIRY PARTICULARS

Customer Requirements

1.1.1	Application	: Induction Hardening
1.1.2	Component	: Bar Shaft
1.1.3	Material specification	: SAE10B38M2
1.1.4	Hardness specification	: 55-59 HRC (Surface Hardness)

1.2 Recommendations

1.2.1	Power source Hardening	: SINAC SM 100/160, Single Output 4-6 kHz 9-13 kHz (X 3 No's)
1.2.2	Handling Machine	: VM 1000 1x3, 7 Servo Axes (Triple Spindle)
1.2.3	Cooling system	: RC 500 Mj/hr-1000 L-400 LPM
1.2.4	Quench Water re-circulating system	: RQ 300 Mj/hr X 3-2000 L-160x3.470
1.2.5	Cycle time for 31 mm dia 769 mm Length Component is	: 27 Sec +15 % tolerance also component cycle time may vary depending on Size and case depth of the component.

(Suhas S Rao)

(Santosh M)

SCOPE OF SUPPLY OF THE PROPOSED MACHINE & PRICING

Accordingly, we have considered the following features in the machine:

Sl.No	Description	Unit	Qty	Technical Specification & make
1.00	Machine (CNC & Electricals)			
	CNC Model Siemens 840 D with UPS	No.	1	Siemens 840 D SL, 7 axes
	Control Panel with Electricals & Panel Air conditioner	No.	1	Rittal
	Panel Air conditioner	No.	1	Suitable
	Distribution Panel with Electricals	No.	1	Rittal
	Electrical Integration & Cabling	No.	1	Various
	RFID for Coils for Coil Identification & Life Monitoring	No.	3	EFD Induction
	Pendants and mounting arm for CNC	No.	1	Suitable
	Sensors, Sensor mountings and Integration	No.	1	IFM / TE
2.00	Handling Machine (Mechanical, Electrical, Pneumatic & Others)			VM 1000 1x3 7 Axes
	Main Frame , Machine Housing & Guards	No.	1	SS Frame
	Vertical servo axis assembly (Z axis) 1.0 Mtr.	No.	1	Hiwin/Rexroth
	Horizontal servo axis assembly (Y axis)	No.	1	Suitable
	Vertical servo axis assembly (Z axis) for tailstock	No.	1	Suitable
	Triple main spindle with SEW geared motor and VFD	No.	1	EFD Induction/SEW
	Carriage & 2D table with sealing arrangement with LVDT for X axis adjustment (All the three Coil are mounted on a Single 2D table so, Individual adjustment of coil in X&Y direction is not possible)	No.	1	EFD Induction
	Cable drag chain to carry output power cables	No.	3	IGUS make
	Triple tailstock with pneumatic top floating centers	No.	1	EFD Induction
	Linear retractable job loading support	No.	1	Festo
	Moving rest with rollers for each shaft and pneumatic forward/backward movement	No.	1	EFD Induction
	Safety light curtain	No.	1	Suitable
	Busbar(3 No's), Rollaway covers, Quench mtg.	No.	1	EFD Induction

	Manual front access door, Machine Safety Guarding and Maintenance Access Door	No.	1	EFD Induction
	Limit switches for access doors	No.	1	Suitable
	Pneumatics, Auto Lubrication, Painting, Electro-Pneumatic Solenoids	No.	1	Festo, GEMU
	Component Length check.	No.	1	EFD Induction
3.00	Machine Accessories			
	Inlet stacker and shaft ejector with poka-yoke for reverse loading (rejection box) with stacking of 200 Nos max.	No.	1	EFD Induction
	Inlet Walking beam	No.	1	EFD Induction
	Component Outlet system (Non Chain type)	No.	1	EFD Induction
	Gantry with X,Y & Z axis for grippers and tilting unit with gripper cluster	No.	1	EFD Induction
	Pick and place arrangement at outlet to feed next machine	No.	1	EFD Induction
	Mist collector with mounting arrangement	No.	1	Span
	Vented hood for the workstation	No.	1	SS
	Rejection box interlocked with CNC for IH machine	No.	1	EFD Induction
	Air & Quench jet washing guns	No.	1	FESTO
	Air Purging after Induction Hardening	No.	1	Suitable
4.00	Power source and accessories			
	Power Source(type, model, frequency, outputs)			
	SINAC SM 100/160, Single output, 4-6 kHz & 9-13 kHz	No.	3	EFD Induction
	Earth fault detection to detect non-heating & protect coil / component	No.	3	EFD Induction
5.00	Cooling and Quench systems & accessories			
	Quench re-circulating system (RQ 300 Mj/Hr X 3-2000 L-160 LPMx3. 470)/Suitable with Heater	No.	1	Grundfoss / Alfa Level
	Cooling water re-circulating system (RC 500 Mj/hr -1000L-400 LPM)	No.	1	Grundfoss / Alfa Level
	Magnetic separator and oil skimmer arrangement	No.	1	SPAN / U-tech
	SS Compact band paper filter	No.	1	SPAN
	Booster pump for coil cooling water circuit	No.	1	Grundfoss
	Intermediate tank (500 L)	No.	1	Grundfoss
	Frame for cooling systems	No.	1	EFD Induction
	Frame for compact band filter and piping	No.	1	EFD Induction
6.00	Integration			

	Integrated piping & instruments	No.	1	SS
	Integrated cabling	No.	1	LAPP
7.00	Process monitoring system			
	(Energy – 3 , Quench Flow – 3 , Quench Temperature-3)	No.	1	EFD Induction
8.00	Process & Set-up tooling			
	Inductors with Quench Ring	No.	3+6	EFD Induction
	Top Bottom Centers	No.	3	EFD Induction
	Inspection gauges for coils (for 1 variety of coil)	No.	1	EFD Induction
	Inspection gauges for coils on machine (for 1 variety of coil)	No.	1	EFD Induction
9.00	Process prove-out & Pre-inspection			
	Component varieties for tests			2 Variety during Pre acceptance at EFD and 5 Variety of shafts at M/s Hyundai WIA
	Productivity & Consistency tests, number of components			300 Nos Per Variety
	Laboratory inspection			First, Middle and Last (For each Spindle, Total – 9 components per variety)
	Customer training			3 days
	Machine inspection			3 days
10.00	Spare Part List (Find below list of spares)	-	-	EFD
11.00	Packing-Base Plate Packing			

PRICE BASIS: FOB, EFD INDUCTION PRIVATE LIMITED, BANGALORE.

PRICE FOR ITEM 1 THRU 11.....: INR 4,67,00,000/-

(Rupees Four Crore Sixty Seven Lakhs Only)

Installation & Commissioning charges additional: INR 3,00,000/-

(Rupees Three Lakhs Only)

Spare Parts List as Follows:

SL.No	Item No.	Description	Qty
1	PEM00012	FUSE	3
2	PEY00032	TRANSISTOR MODULE - IGBT	7
3	IEN00341	ELECTRONIC CARD – SMPS	2
4	IEN00338	ELECTRONIC CARD – POWS	1
5	SIFC0021	ELECTRONIC CARD – ASMO	1
6	SIFC0019	ELECTRONIC CARD – POCC	1
7	IEN00338	ELECTRONIC CARD – POFR	2
8	SIFC0020	ELECTRONIC CARD – MIFC	1
9	SIFC0016	ELECTRONIC CARD – VCPM	1
10	SIFC0017	ELECTRONIC CARD – MGDC	2
11	SIFC0012	ELECTRONIC CARD – MGDA	1
12	IEN00698	VARISTOR	1
13	FIHH0016	HAND HELD TRANSFORMER 1200	1
14	PEH00026	TRANSFORMER THREE PHASE	1
15	SICM0030	TRANSISTOR CONTROL CABLE	4
16	IOT00680	COOLING FAN	1
17	IOT01384	ISOLATION AMPLIFIER	1
18	IOT00693	DISPLAY FOR ISOLATION AMPLIFIER	1

(Suhas S Rao)

(Santosh M)

DELIVERY AND PAYMENT CONDITIONS

Please mention below terms in your purchase order:

“ Termination for convenience (cancellation) of a Purchase Order

In the case of a Purchase Order for the delivery of customized equipment and subject to the conditions below, EFD Induction allows for termination for convenience by the Purchaser.

In such case, EFD Induction shall be entitled to invoice to the Purchaser the work carried out and materials or services received to the date for the execution of the purchase order as well as the work, materials or services already ordered with suppliers to the extent that these cannot be cancelled. In addition, EFD Induction shall be entitled to a compensation for profit equal to 15% of the balance between the aforesaid amount and the Purchase Price.

To avoid the determination of the degree of advancement or the works contracted and to limit the Purchaser's risk, EFD Induction prefers to enter into a cancellation scheme to apply in the case of termination for convenience. Such scheme shall be agreed in advance and become integral part of the Purchase Order.

Upon payment of the aforesaid amounts, any work carried out or materials received for the Purchaser Order shall become property of the Purchaser. EFD Induction shall act in good faith as regards the buy-back for re-use or re-sale of work or materials that the Purchaser cannot use.

Regarding the delivery of standard products in the context of purchase agreements or work performance contracts, termination for convenience by the Purchaser remains at the discretion of EFD Induction. When a termination is accepted, EFD Induction shall be entitled to compensation as stated above.”

1. **PRICE BASIS:** The above price is on Ex-Works Bangalore basis with base plate packing. All taxes and duties will be extra at actuals. Freight & transit insurance chargeable to Your account at actuals. IGST @18% is applicable currently.
2. **DELIVERY:** 9.5 Months from the date of technically and Commercially clear LOI/PO receipt at EFD Inductions India. Also any delay in DAP approval from Customer, providing detailed component drawing, arranging for trial parts and delay in visiting for Pre acceptance will cause proportionate delay in project and the delivery date to be extended accordingly by Customer.
4. **VALIDITY:** Thirty days from the date of this offer.
5. **TERMS OF PAYMENT:**

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- 30% Advance payment against corporate guarantee.
- 50% With GST against delivery at M/s Hyundai Wia, India.
- 20% Against Installation & Commissioning at M/s Hyundai Wia, India.

6. **PRICE VARIATION CLAUSE:** The equipment contains substantial imported items, costs of which are dependent on exchange rate of Indian currency. Our quoted price is based on exchange rate of Rs. 77/EURO. Any fluctuation in the exchange rate unfavorable to the Indian currency will be charged at actuals.
7. **PRE - ACCEPTANCE TEST:** Prices include the demonstration of process for maximum of Two varieties of components. Work pieces must be identical to the parts which will be used for final production at buyer's works. Test pieces to be demonstrated must be available in sufficient quantity at our works at least 12 weeks before the agreed delivery date. If receipt of test pieces is delayed, scheduled delivery may take place without demonstration or written confirmation of the buyer. The same clause regarding delivery will apply if clearance is delayed even after successful process demonstration.
8. **FINAL ACCEPTANCE:** Price includes the demonstration of the process for maximum of Five varieties of components. The varieties to be proved during final Acceptance should be same as components provided in the Pre-acceptance.
9. **PAINTING:** The equipment will be painted in your standard.
10. **WARRANTY:** Provided our instructions are put into practice, 12 months from the date of Commissioning or 18 Months from the date of Dispatch whichever comes first. Warranty for CNC 18 Months from the date of Commissioning or 24 Months from the date of dispatch whichever comes first. If you use any spurious parts in our machine, then immediately warranty will be void and EFD is not responsible for any breakdowns.
11. **RETENTION OF TITLE:** Title to all goods will be retained with us till our each and every claim against the buyer originating in the course of the contract has been fully satisfied.
12. **COPY RIGHT:** The specification and technology contained in our proposal is the property of EFD. Passing on this information to third parties as well as unauthorized utilization by the buyer himself are both not permitted.
13. **SUBSEQUENT ALTERATIONS TO ORDER:** If the order is altered, the price and delivery time may be revised.

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14. **DISPUTES:** In accordance with UN 188 A with addendum of 1987.
15. **CANCELLATION:** Actual project costs up to the date of cancellation is to be paid by the customer, plus 18% of total contract value as a compensation for production re-adjustment costs and loss of project income.

(Suhas S Rao)

(Santosh M)

M/s HYUNDAI-WIA (INDIA) PVT. LTD.

**SALES PROPOSAL
FOR
TULIP INDUCTION HARDENING**



EIN112384

TULIP HARDENING AND MACHINE

APPLICATION:

Induction Hardening of bell and stem portions of Tulip in a single sequence.

COMPONENTS TO BE ACCOMODATED IN MACHINE:

Machine will be designed to accommodate only Tulip components mentioned in the RFQ XL document provided "180427_India 5th project" Without Induction Tempering.

CYCLE TIME:

The cycle time is 25 seconds (FTF).

Please Note: The estimated cycle time is calculated theoretically and is subject to trials at our end. Please consider a tolerance of +15% on the above estimated cycle time. Also the cycle time may vary as per the geometry of the component.

We would like to draw your attention to the following points of our offer:

- 1) *This machine is dedicated to run Tulip component only, we will not be able to run O-Race component in this machine.*
- 2) *We have attached your RFQ "180427_India 5th project" with our comments.*
- 3) *We have Removed Tempering from our Scope.*
- 4) *We will provide a Cylinder and Chute to push the components one after the other to Electrical heater only (This will be similar to our earlier supply to Hyundai Wia India Plant). After tempering the component have to be fed to our Antirust Appling machine. The conveyor required for transporting component from Tempering to Antirust machine is not in our Scope of Supply.*
- 5) *This proposed machine is not capable of Hardening Female bore for Tulip.*
- 6) *The machine is designed to handle the components of Max. Dia. Of 130 mm and Max. Length of 250 mm.*
- 7) *We have not included 'Laser marking' in our scope of supply.*
- 8) *We can have a dry run of the machine for 2 Day, machine will be operating 8 Hrs per day without Alarm before Pre acceptance.*
- 9) *"Penalty will be borne by supplier, if supplier delay each steps as approval, design, built machine."*

EFD Comment: We accept 0.5% per week delay and to the maximum of 5% of contract value for delivery.

Note:

Delivery penalty will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai-WIA., some of the crucial/Major activities are listed below:-

- i) Delay in DAP approval.
- ii) Delay in Advance Payment.
- iii) Delay in providing Samples for trials.
- iv) Delay in visiting for inspection and dispatch clearance.
- V) Delay in providing correct drawing of component to design tooling.

10) "Work Quality Chart":

순	Control items	Control standard	Remarks
1	Surface hardness	- After Queching : 59 ~ 64(HRc) - After Tempering : 58 ~ 64(HRc) (Recommended 58-61 HRC)	Less 2~3 HRC after tempering
2	O.P.D distortion	- Max : 0.02mm (Serr. Area within 0.01mm top and bottom)	
3	Structure	- BELL : ALL Martensite - STEM : Pearlite + Ferrite (center $\Phi 3$ area)	
4	Groove distortion	- Top/middle/bottom should be less than 0.03mm	

EFD Comments: We can reproduce the same results which we have delivered to M/s. Hyundai-WIA (India). We have selected Power sources with little higher power capacity, so that you can have the flexibility to conduct different trial to arrive at the correct result to pass the Mechanical and Torsional testing of components.

11) "TOOL SIZE and SET UP should be monitoring type".

EFD Comments: We have included coil RFID for coil identification and Coil life monitoring.

12) We have considered a critical spare parts worth INR 23, 81, 892/-

13) During trials at customer site, Customer to provide lab support to conduct component inspection.

14) We will provide all the documents in English Language only.

15) We have included 2 Nos of Stem coil and 16 Nos of Bell coil for Eddy current checking systems in our Tulip machine.

16) We need water from Customer's cooling tower/water chiller at 25-30°C, 2.5-4 Bar with 1000 LPM flow. Supply of cooling capacity of 25 TR is not sufficient for Tulip machine we need 67 TR.

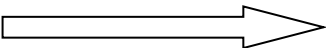
MACHINE CONFIGURATION:

Handling Machine : LHM
 Power source : SINAC 200/320 SM, IGBT Based, Double Output,
 Maximum Output Power 320 kW in intermediate duty,
 Output Frequency 20-30 kHz for Bell and 9-13 kHz for Stem.

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Quench re-circulating : RQ 350x2-2000-300x2/Suitable 7,5 bar or suitable system
Cooling water re-circulating : RC 350-1000-250/Suitable 7,5 bar or suitable system
Process tooling : Suitable for 9 Varieties of Tulip

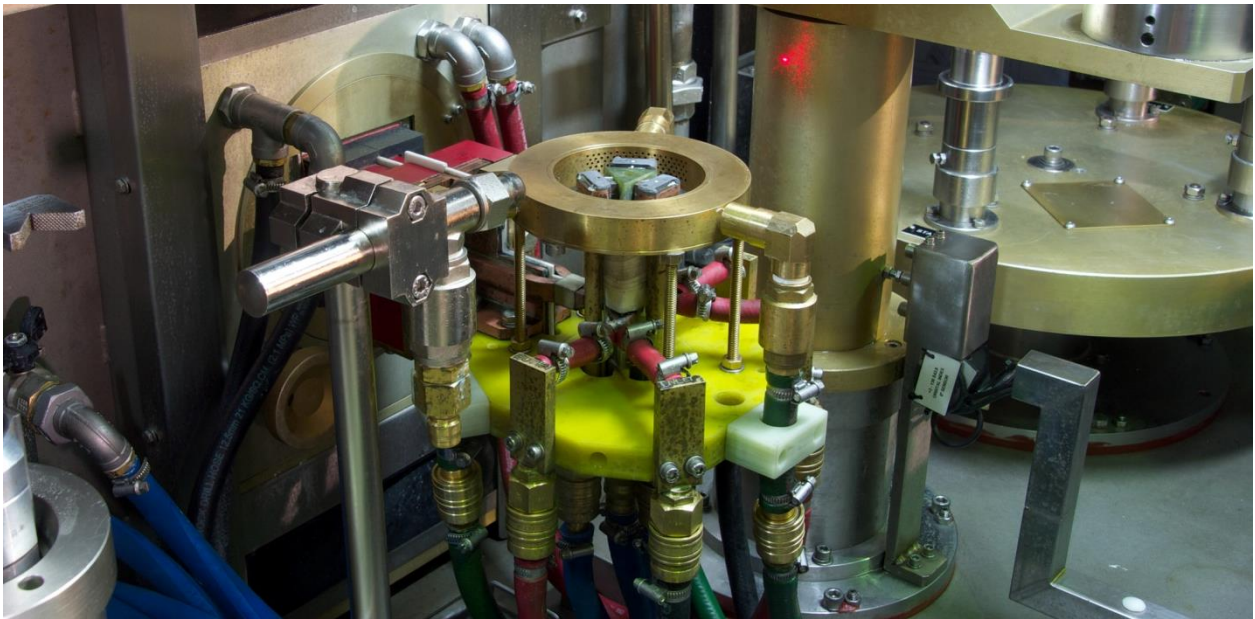
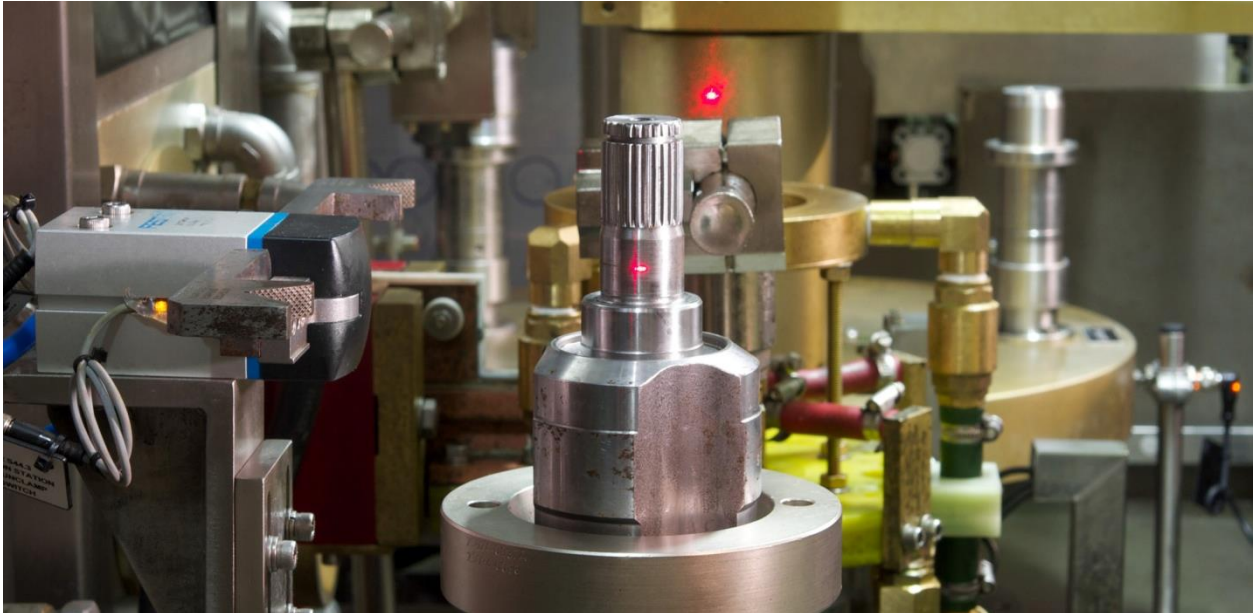
PROCESS SEQUENCE:



Inlet Stack	Washing	Orientation	Bell IH	Stem IH	Drying	ECT	Dot matrix Marking	Anti rust application	Out Let Stacker
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Please find herewith some photos of Similar machine for you reference only:-





SCOPE OF SUPPLY:

Sl.No	Item / Assembly	Unit	Qty	Technical Specification & make
1.00	Machine (CNC & Electricals)			
1.01	Siemens 840D SL with UPS for CNC	Set	1	SIEMENS 840D SL 4 Axis
1.02	Control & Distribution Panel & Electricals (Rittal Cabinets)	Set	1	Rittal Panels
1.03	Earth fault detection for Only Hardening Machine	Set	2	EFD
1.04	CNC Screen mounting	Set	1	Suitable
1.05	Panel air conditioner	Set	1	Suitable
1.06	RFID for coil identification and coil life monitoring	No.	2	EFD
1.07	Analog module for integration of LVDT	Set	2	Suitable
1.08	Variable Frequency Drive to vary component rotation speed	No.	1	Suitable
1.09	Safety Switch for main door	Set	1	Suitable
1.10	Color sensors to prevent double hardening in stem station only	Set	1	Suitable
1.11	Proximity Sensors, Three tier lamp, Poka-yoke, Electrical Controls & Indicators	Set	1	IFM / TE
2.00	Hardening Line (Mechanical, Electrical, Pneumatic & Others) ,			SPM
2.01	Inlet and Outlet stacker with 200 No's holding capacity	Set	1	EFD
2.02	Component washing unit with re-circulating water system with filtration, oil skimmer, heater, air blowing and walking beam for job transfer	Set	1	EFD
2.03	Main Hardening machine			
2.04	Main Frame (SS Basin) , Machine Housing & Guards (Linear Frame)	Set	1	EFD
2.05	LVDT for component height checking	Set	1	EFD
2.06	Component orientation Assembly	Set	1	EFD
2.07	Z-axis for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Suitable
2.08	X-axis, for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Suitable
2.09	Z-axis, Suitable for Component up and down movement in stem station (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Suitable
2.10	3-Jaw Chuck to pick the component from orientation station to heating station(Please note:- The three jaw chuck will not rotate in Tulip Machine)	No.	1	Jaws/Suitable

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2.11	Component Intermediate station	No.		EFD
2.12	Additional gripper to transfer the component Intermediate station to Stem Station	No.	1	EFD
2.13	SEW AC geared motor for component rotation at stem station and rotation assembly	No.	1	SEW
2.14	2D table with sealing arrangement for Stem Coil with LVDT for X & Y axis adjustment	No.	1	EFD
2.15	2D table with sealing arrangement for Bell Coil with LVDT in X & Y adjustment	No.	1	EFD
2.16	Feed table (Feed table movement is by Servo motor) for Stem Station with work Piece Lifter	No.	1	EFD
2.17	Tailstock for stem station	No.	1	EFD
2.18	Quenching arrangement for stem station	No.	1	EFD
2.19	Idle station with additional quenching arrangement	Set	1	EFD
2.20	Cable drag chain for Stem station only	Set	1	EFD
2.21	Busbar with Sealing	No.	2	EFD
	Automation (Component Handling for Hardening Station)			
2.22	Input pick & place arrangement with pneumatic z-axes and x-axes (washing to Orientation station)	No.	1	EFD & FESTO
2.23	Output pick & place arrangement (shuttling table to outlet conveyor)	No.	1	EFD & FESTO
2.24	Rejection bin suitable for Tulip with interlock to CNC	Set	1	EFD
2.25	Manual door, Machine Safety fence with safety interlocks and Maintenance Access Door	No.	1	EFD/CTEKNIC EUCHNER
2.26	Safety Door Locks	No.	1	Suitable
2.28	Air blow station before ECT and marking	No.	1	EFD
2.29	Eddy current Structural test Station with automation for Bell and stem checking with (Eddy Coils: Stem 1set = 2 No, Bell 8 set = 16 No)	No.	1	EFD
2.30	Dot matrix marking of component	No.	1	EFD
2.31	Pick and place walking beam for transporting the Component for ECT and marking	No.	1	EFD
2.32	Anti-rust spray unit with, spraying nozzles, chamber and collection tank, Air blowing with Walking beam to transport the component.	NO.	1	EFD
3.00	Machine Accessories			
3.01	Pneumatic elements and solenoids	Set	1	FESTO
3.02	Auto lubrication, circuit and pump, collection trays	No.	1	Cenlube
3.03	Vented hood to connect fume extractor	No.	1	EFD

3.04	Mist collector with mounting arrangement one for IH	No.	1	Span
3.05	Machine painting, labeling	Set	1	As per WIA St
3.06	SS Compact paper band filter	Set	1	SPAN
3.07	Air & Quench jet washing guns	Set	1	EFD
3.08	3-tier lamps, machine indications and controls, machine lamp	Set	1	Various
4.00	Power source and accessories			
	Power Source(type, model, frequency, outputs)			
4.01	SINAC 200/320 kW SM double output, 20-30 kHz for Bell, 9-13 kHz for male stem,	No.	1	EFD
5.00	Cooling and Quench systems & accessories			
5.01	Cooling System	Set	1	(RC 350-1000-250)/Suitable
5.02	Quench re-circulating system (Individual pumps & heat exchangers for stem and bell) with oil skimmer	Set	1	(RQ 350x2-2000-300x2)/Suitable
5.03	Magnetic separator	Set	1	Suitable
5.04	Booster pump for cooling water circuit (High Pressure)	Set	1	Grundfoss
5.05	Intermediate tank + Collection tank after filtration with circuit	Set	1	EFD
5.06	Frame for cooling systems with integral ladder - guarding	Set	1	EFD
6.00	Integration			
6.01	Integrated piping & instruments	Set	1	SS
6.02	Electrical Integration & Cabling	Set	1	EFD
7.00	Process monitoring system			
7.01	Energy Monitoring System	No.	2	EFD
7.02	Quench Flow Monitoring System	No.	9	IFM
7.03	Quench Temperature Monitoring System	No.	1	EFD
8.00	Process tooling for Tulip (9 Varieties of Tulip)			
8.01	Stem coil	No.	6+6	EFD
8.02	Bell Coil	No.	8+8	EFD
8.03	Orientation fixture	No.	9+9	EFD
8.04	Fixture on feed table	No.	27+27	EFD
8.05	Coil setting mandrels	No.	14	EFD
8.06	Spare top center for Stem station	No.	1	EFD
9.00	Process Prove Out and Consistency Test			

9.01	Component varieties for tests			2 variety at EFD for Pre acceptance & 9 variety prove out at M/s Hyundai WIA India plant
9.02	Productivity & Consistency tests, number of components			150 No's.each Variety
9.03	Laboratory inspection			First, Middle & Last
9.04	Customer training			3 days
9.05	Machine inspection			3 days
10.00	Spare Part (Please refer the List below)			EFD
11.00	Packing			

Note:

1. For cooling re-circulating system, you have to provide cooling water from your cooling tower to the heat exchangers at a temperature range of 28-32°C at a pressure of 4 to 6 bar.
2. Customer to unload the machine and position as per the layout provided by EFD at its site and
3. Provide all the utilities as per the requirements of the machine.
4. Customer to provide component cutting facility during trials at customer location.
5. Important note on Price Breakup:- a) Since this is a project execution, the price break-up provided above is just notational as it is not possible to break-up the cost of engineering, training costs, costs towards warranty etc.
b) The Price breakup is just for your understanding and we request you not to consider the above prices as benchmark for any future order placed on us.

PRICING:

For items 1.00 through 12.00 listed above including Packing and loading on to truck at EFD Induction India.

Freight charges and Transit Insurance to be arranged by customer:-

Basic Machine price including Packing and Loading on to truck:- :-

₹ 4,99,40,000/-

(Rupees Four Crore Ninety Nine Lakhs Forty Thousand Only)

Installation & Commissioning charges extra:- ₹ 3,60,000/-

(Rupees Three Lakhs Sixty Thousand Only)

Spare Parts List as Follows:

SL.No	Item No.	Description	Qty
1	PEM00012	FUSE	3
2	PEY00032	TRANSISTOR MODULE - IGBT	7
3	IEN00341	ELECTRONIC CARD – SMPS	2
4	IEN00338	ELECTRONIC CARD – POWS	1
5	SIFC0021	ELECTRONIC CARD – ASMO	1
6	SIFC0019	ELECTRONIC CARD – POCC	1
7	IEN00338	ELECTRONIC CARD – POFR	2
8	SIFC0020	ELECTRONIC CARD – MIFC	1
9	SIFC0016	ELECTRONIC CARD – VCPM	1
10	SIFC0017	ELECTRONIC CARD – MGDC	2
11	SIFC0012	ELECTRONIC CARD – MGDA	1
12	IEN00698	VARISTOR	1
13	FIHH0016	HAND HELD TRANSFORMER 1200	1
14	PEH00026	TRANSFORMER THREE PHASE	1
15	SICM0030	TRANSISTOR CONTROL CABLE	4
16	IOT00680	COOLING FAN	1
17	IOT01384	ISOLATION AMPLIFIER	1
18	IOT00693	DISPLAY FOR ISOLATION AMPLIFIER	1

MACHINE FEATURES:

1. POWER SOURCE MODEL (SINAC SM)

The power source offered (SINAC 200/320 SM) is of transistorized (IGBT) design with peak output power of 320 kW intermittent duty and 200 kW in continuous duty. The power source is a double output version. The output has an operating frequency band is 20-30 kHz for Bell, 9-13 kHz for Stem hardening.

Item	Qty	Description
1		Power Source
1.1	1	Transistorised Frequency Converter (series compensated) Type SINAC 200 / 320 SM, single output Max. Output power (intermittent duty) 320 kW Output power (continuous) 200 kW Frequency range 20-30 (Bell) & 9-13 kHz (Stem) Features and Modules • Compact cabinet for indoor operation (IP 54 protection) • Main input terminal • Diode rectifier • Converter with IGBT-transistors • Output capacitor module • Electronic control device • Indirect cooling water system • External or internal power pre-selection/ regulation • Operating panel as plug - in unit with indicating instruments and transparent protection panel • LED single fault display (alarm card)
1.2	2	Water Cooled Hand Held Output Transformer.
1.3	2	Set of inductor connecting brackets, quick clamping type (manual) with groove guide for precise inductor clamping

EFD Induction Private Limited

2. RE-CIRCULATING TYPE COOLING SYSTEMS FOR QUENCH MEDIUM AND COOLING WATER

- 2.1 1 **Cooling Unit for Power Source, Output Transformer, Busbar and Coils**
(heat exchanger principle)
Type: (RC 350-1000-250)/Suitable

Water tank in stainless steel, strainer, circulating pump with motor, plate type heat exchanger, pressure gauges, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

Cooling capacity	350 MJ
Tank capacity	1000 L
Pump flow rate	250 L/min.
Pump pressure	7,5 bar

- 2.2 1 **Cooling Unit for Quench Medium**
(Heat exchanger principle)
Type: (RQ 350x2-2000-300x2)/7-5 or suitable

Water tank in stainless steel, Compact paper band filter, with strainers, circulating pump with motor, Pressure gauges, plate type heat exchanger, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

Cooling capacity	350 MJ X 2
Tank capacity	2000 L
Pump flow rate for	300 L/min X2
Pump pressure	7,5 bar

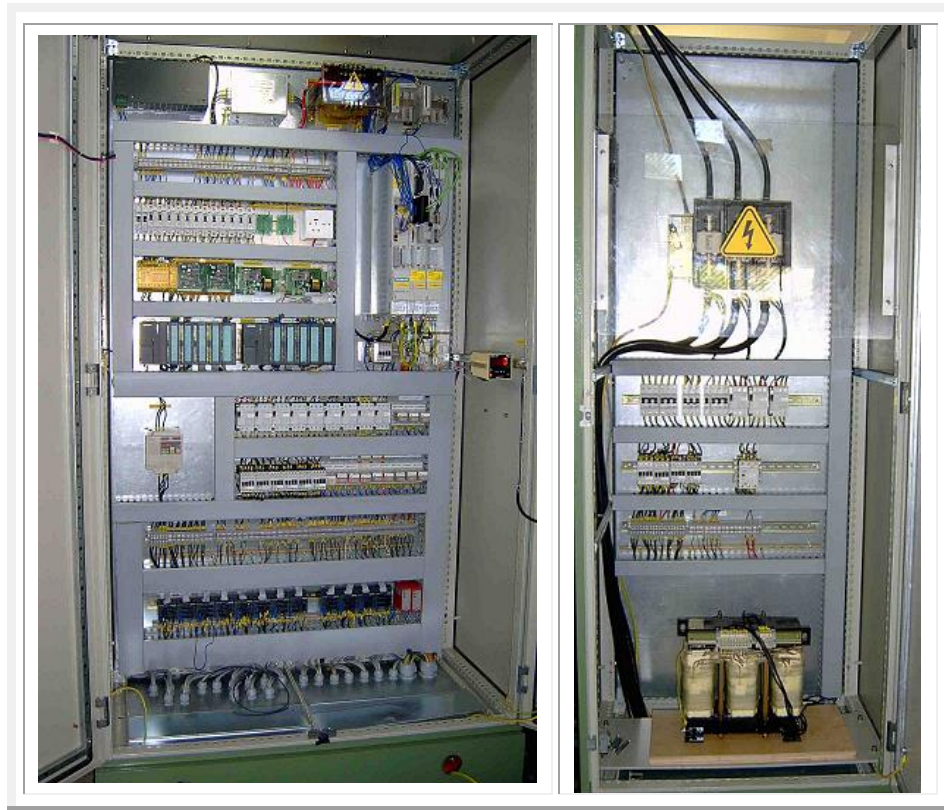
Please take notice:

The re-circulating systems must be provided with cooling water, max. 32°C, min. 4 to 6 bar pressure, flow rate equivalent to the stated circulating pumps.

Accessories

- 2.3 1 Platform to assemble cooling units on top of each other with integral ladder

3. CONTROL PANEL HOUSING CNC DRIVES, ELECTRICAL SWITCH GEAR AND DISTRIBUTION PANEL HOUSING MAINS AND FUSES



4. PNEUMATIC CONTROL PANEL





COMMERCIAL TERMS:

We request you to include the below statements in you PO.

“ Termination for convenience (cancellation) of a Purchase Order

In the case of a Purchase Order for the delivery of customized equipment and subject to the conditions below, EFD Induction allows for termination for convenience by the Purchaser.

In such case, EFD Induction shall be entitled to invoice to the Purchaser the work carried out and materials or services received to the date for the execution of the purchase order as well as the work, materials or services already ordered with suppliers to the extent that these cannot be cancelled. In addition, EFD Induction shall be entitled to a compensation for profit equal to 15% of the balance between the aforesaid amount and the Purchase Price.

To avoid the determination of the degree of advancement or the works contracted and to limit the Purchaser's risk, EFD Induction prefers to enter into a cancellation scheme to apply in the case of termination for convenience. Such scheme shall be agreed in advance and become integral part of the Purchase Order.

Upon payment of the aforesaid amounts, any work carried out or materials received for the Purchaser Order shall become property of the Purchaser. EFD Induction shall act in good faith as regards the buy-back for re-use or re-sale of work or materials that the Purchaser cannot use.

Regarding the delivery of standard products in the context of purchase agreements or work performance contracts, termination for convenience by the Purchaser remains at the discretion of EFD Induction. When a termination is accepted, EFD Induction shall be entitled to compensation as stated above.”

1.0 Price Terms:

- ☐ Ex-works without packing and forwarding
- ☒ Ex-works with packing and forwarding
- ☐ Free delivery up to customer's site
- ☐ Free delivery to transport office
- ☐ Free delivery to nearest port

All taxes and duties will be extra at actual and at rates prevailing at the time of despatch.

2.0 Packing Type:

EFD Induction Private Limited

- ☒ Wooden platform and machine covered with bubble sheet
- ☐ Carton box packing with nylon straps
- ☐ Complete Wooden Box Packing for transport by road / rail
- ☐ Sea-worthy packing

3.0 Installation & Commissioning Charges:

- ☒ Included ☐ Not Included

4.0 Scope of other charges:

- | | | |
|------------------------|--|--|
| Transportation Charges | ☐ Included ☐ Not | ☒ Customer's Scope |
| Transit Insurance | ☐ Included ☐ Not | ☒ Customer's Scope |
| Others | ☐ Included ☐ Not | ☒ Customer's Scope |

5.0 Delivery Period: 9.5 Months

The delivery period is counted from the date of our Order Acceptance. The Order Acceptance will be sent within Five days of EFD receiving the printed copy of the Purchase Order/LOI. The order acceptance is to confirm preliminary acceptance of the order. However, if there are any deviations found in order after detailed contract review it will be addressed in a separate correspondence from EFD. If due to such changes, the delivery date has to be extended it shall be done as per EFD's request.

6.0 Penalty Clause: - LD charges 0.5% per week & up to maximum 5% of basic value. Penalty

Applicable after two weeks of Delivery date.

Note:

LD will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai WIA, some of the crucial / Major activities are listed below:-

- i) Delay in DAP approval.
- ii) Delay in advance Payment.
- iii) Delay in providing samples for trials.
- iv) Delay in visiting for Inspection and dispatch clearance.
- V) Delay in providing correct drawing of component to design tooling.

7.0 Payment Terms:

- 30% Advance payment against corporate guarantee.
 - 50% With GST against delivery at M/s Hyundai Wia, India.
- EFD Induction Private Limited



20% Against Installation & Commissioning at M/s Hyundai Wia, India.

8.0 VALIDITY: Thirty days from the date of this offer.

9.0 PRE - ACCEPTANCE TEST:

Prices include the demonstration of process for maximum of two variety of component. Work pieces must be identical to the parts which will be used for final production at buyer's works. Test pieces to be demonstrated must be available in sufficient quantity at our works at least 12 weeks before the agreed delivery date. If receipt of test pieces is delayed, scheduled delivery may take place without demonstration or written confirmation of the buyer. The same clause regarding delivery will apply if clearance is delayed even after successful process demonstration.

10.0 FINAL ACCEPTANCE:

Price includes the demonstration of the process for maximum of 9 variety of Tulip component. Additional prove-outs will be undertaken at extra charges.

11.0 WARRANTY:

12 months from the date of Commissioning or 18 Months from the date of Dispatch whichever comes first. Warranty for CNC 18 Months from the date of Commissioning or 24 Months from the date of dispatch whichever comes first. The warranty does not cover consumables, toolings and other safety elements.

Usage of any spurious parts in EFD machine by the customer, then immediately warranty will be void and EFD is not responsible for any breakdowns.

12.0 RETENTION OF TITLE:

Title to all goods will be retained with us till our each and every claim against the buyer originating in the course of the contract has been fully satisfied.

13.0 COPY RIGHT:

The specification and technology contained in our proposal is the property of EFD. Passing on this information to third parties as well as unauthorised utilisation by the buyer himself are both not permitted.

14.0 SUBSEQUENT ALTERATIONS TO ORDER:

If the order is altered, the price and delivery time may be revised.

EFD Induction Private Limited



15.0 DISPUTES:

In accordance with UN 188 A with addendum of 1987.

16.0 CANCELLATION:

Actual project costs up to the date of cancellation is to be paid by the customer, plus 18% of total contract value as a compensation for production readjustment costs and loss of project income

***“Any deviations to the above terms are valid only
if EFD has accepted in writing prior to order release or during order acceptance.”***

M/s HYUNDAI-WIA (INDIA) PVT LTD
SALES PROPOSAL
FOR
OUTER RACE HARDENING



EIN112385

OUTER RACE INDUCTION HARDENING MACHINE

APPLICATION:

Induction Hardening of bell and stem portions of O-Race in a single sequence.

COMPONENTS TO BE ACCOMODATED IN MACHINE:

Machine will be designed to accommodate the range of O-Race components mentioned in the RFQ XL document provided "180427_India 5th project" without Induction tempering.

CYCLE TIME:

The cycle time is 25 seconds (FTF).

Please Note: The estimated cycle time is calculated theoretically and is subject to trials at our end. Please consider a tolerance of +15% on the above estimated cycle time. Also the cycle time may vary as per the geometry of the component also.

EFD Induction Private Limited

We would like to draw your attention to the following points of our offer:

- 17) This machine is dedicated to run O-Race component only, we will not be able to run Tulip component in this machine.
- 18) We have attached your RFQ "180427_India 5th project" with our comments.
- 19) We have Removed Tempering from our Scope.
- 20) We will provide a Cylinder and Chute to push the components one after the other to Electrical heater only (This will be similar to our earlier supply to Hyundai Wia India Plant). After tempering the component have to be fed to our Antirust Applying machine. The conveyor required for transporting component from Tempering to Antirust machine is not in our Scope of Supply.
- 21) This proposed machine is not capable of Hardening Female bore for Tulip.
- 22) The machine is designed to handle the components of Max. Dia. Of 130 mm and Max. Length of 250 mm.
- 23) We have not included 'Laser marking' in our scope of supply.
- 24) We can have a dry run of the machine for 2 Days, machine will be operating 8 Hrs. per day without Alarm before Pre acceptance.
- 25) "Penalty will be borne by supplier, if supplier delay each steps as approval, design, built machine."

EFD Comment: We accept 0.5% per week for delay and to the maximum of 5% of contract value for delivery.

Note:

Delivery penalty will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai-WIA., some of the crucial/Major activities are listed below:-

- i) Delay in DAP approval.
- ii) Delay in Advance Payment.
- iii) Delay in providing Samples for trials.
- iv) Delay in visiting for inspection and dispatch clearance.
- v) Delay in providing correct drawing of component to design tooling.

26) "Work Quality Chart":

순	Control items	Control standard	Remarks
1	Surface hardness	- After Queching : 59 ~ 64(HRc) - After Tempering : 58 ~ 64(HRc) (Recommended 58-61 HRC)	Less 2~3 HRC after tempering
2	O.P.D distortion	- Max : 0.02mm (Serr. Area within 0.01mm top and bottom)	
3	Structure	- BELL : ALL Martensite - STEM : Pealite + Ferrite (center Φ3 area)	
4	Groove distortion	- Top/middle/bottom should be less than 0.03mm	

EFD Comments: We can reproduce the same results which we have delivered to M/s. Hyundai-WIA (India). We have selected Power sources with little higher power capacity, so that you can have the flexibility to conduct different trial to arrive at the correct result to pass the Mechanical and Torsional testing of components.

27) *“TOOL SIZE and SET UP should be monitoring type”.*

EFD Comments: We have included coil RFID for coil identification and Coil life monitoring.

28) *We have considered a critical spare parts worth INR 23, 81, 892/-*

29) *During trials at customer site, Customer to provide lab support to conduct component inspection.*

30) *We will provide all the documents in English Language only.*

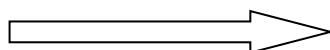
31) *We have included 2 Nos of Stem coil and 10 Nos of Bell coil for Eddy current checking systems in our outer race machine.*

32) *We need water from Customer's cooling tower/water chiller at 25-30°C, 2.5-4 Bar with 1000 LPM flow. Supply of cooling capacity of 25 TR is not sufficient for outer race machine we need 67 TR.*

MACHINE CONFIGURATION:

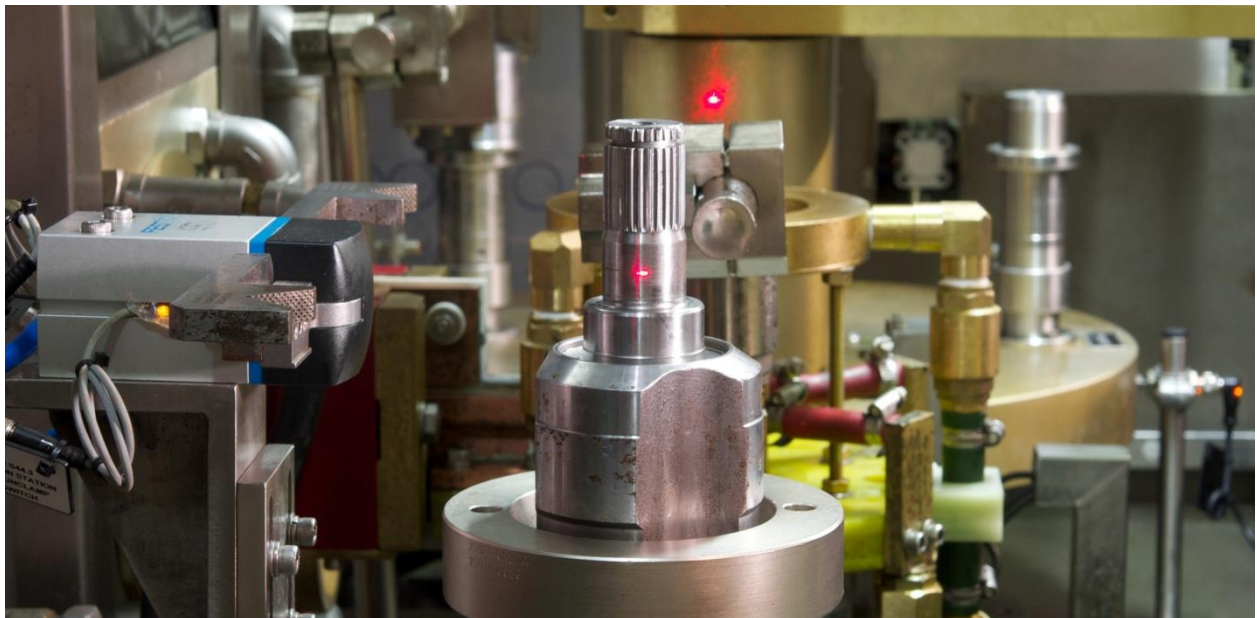
Handling Machine	:	LHM
Power source	:	SINAC 200/320 SM, IGBT Based, Double Output, Maximum Output Power 320 kW in intermediate duty, Output Frequency 20-30 kHz for Bell and 9-13 kHz for Stem.
Quench re-circulating system	:	RQ 350x2-2000-300x2/Suitable 7,5 bar or suitable
Cooling water re-circulating	:	RC 350-1000-250/Suitable 7,5 bar or suitable system.
Process tooling	:	Suitable for 5 Varieties of FOR

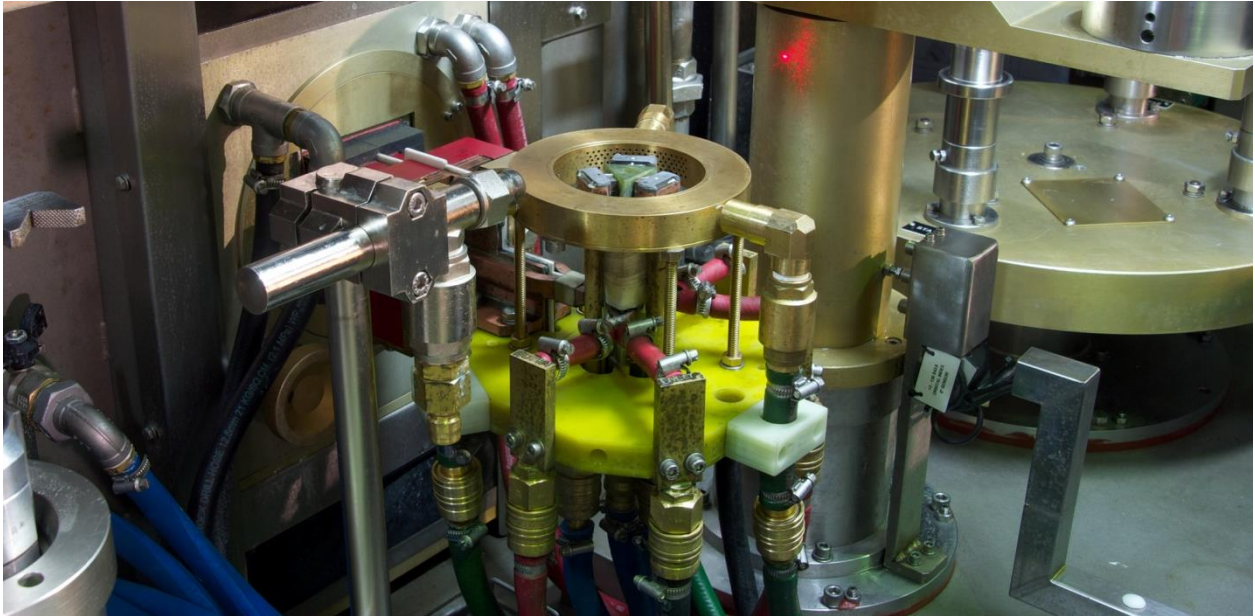
PROCESS SEQUENCE:



Inlet Stack	Washing	Intermediate station	Bell IH	Stem IH	Drying	ECT	Dot matrix Marking	Anti rust application	Out Let Stacker
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Please find herewith some photos of Similar machine for you reference only:-





SCOPE OF SUPPLY:

Sl.No	Item / Assembly	Unit	Qty	Technical Specification & make
1.00	Machine (CNC & Electricals)			
1.01	Siemens 840D SL with UPS for CNC	Set	1	SIEMENS 840D SL 4 Axis
1.02	Control & Distribution Panel & Electricals (Rittal Cabinets)	Set	1	Rittal Panels
1.03	Earth fault detection for Only Hardening Machine	Set	2	EFD
1.04	CNC Screen mounting	Set	1	Suitable
1.05	Panel air conditioner	Set	1	Suitable
1.06	RFID for coil Identification and Coil Life monitoring	Set	2	EFD Induction
1.07	Analog module for integration of LVDT	Set	2	Suitable
1.08	Variable Frequency Drive to vary component rotation speed	No.	1	Suitable
1.09	Safety Switch for main door	Set	1	Suitable
1.10	Color sensors to prevent double hardening in stem station only	Set	1	Suitable
1.11	Proximity Sensors, Three tier lamp, Poka-yoke, Electrical Controls & Indicators	Set	1	IFM / TE
2.00	Hardening Line (Mechanical, Electrical, Pneumatic & Others) ,			SPM
2.01	Inlet and Outlet stacker with 200 No's holding capacity	Set	1	EFD
2.02	Component washing unit with re-circulating water system with filtration, oil skimmer, heater, air blowing and walking beam for job transfer	Set	1	EFD
2.03	Main Hardening machine			
2.04	Main Frame (SS Basin) , Machine Housing & Guards (Linear Frame)	Set	1	EFD
2.05	LVDT for component height checking	Set	1	EFD
2.06	Z-axis for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Hiwin/Rexroth
2.07	X-axis, for Gripper (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Hiwin/Rexroth
2.08	Z-axis, Suitable for Component up and down movement in stem station (Ball screw / Nut, Steander / L.M Guides, Coupling)	Set	1	Hiwin/Rexroth
2.10	3-Jaw Chuck to pick the component from height checking station to heating station with Component	No.	1	Jaws/Suitable

	rotation for Bell Hardening.			
2.11	Component Intermediate station	No.	1	EFD
2.12	Additional gripper to transfer the component Intermediate station to Stem Station	No.	1	EFD
2.13	SEW AC geared motor for component rotation at stem station and rotation assembly	No.	1	EFD
2.14	2D table with sealing arrangement for Stem Coil with LVDT for X & Y axis adjustment	No.	1	EFD
2.15	2D table with sealing arrangement for Bell Coil with LVDT in X & Y adjustment	No.	1	EFD
2.16	Feed table (Feed table movement is by Servo motor) for Stem Station with work Piece Lifter	No.	1	EFD
2.17	Tailstock for stem station	No.	1	EFD
2.18	Quenching arrangement for stem station	No.	1	EFD
2.19	Idle station with additional quenching arrangement	Set	1	EFD
2.20	Cable drag chain for Stem station only	Set	1	EFD
2.21	Busbar with Sealing	No.	2	EFD
	Automation (Component Handling for Hardening Station)			
2.22	Input pick & place arrangement with pneumatic z-axis and x-axis (washing to height checking station station)	No.	1	EFD & FESTO
2.23	Output pick & place arrangement (shuttling table to outlet conveyor)	No.	1	EFD & FESTO
2.24	Rejection bin suitable for Outer race with interlock to CNC	Set	1	EFD
2.25	Manual door, Machine Safety fence with safety interlocks and Maintenance Access Door	No.	1	EFD/CTEKNIC EUCHNER
2.26	Safety Door Locks	No.	1	Suitable
2.27	Air blow station before ECT and marking	No.	1	EFD
2.28	Eddy current Structural test Station with automation for Bell and stem checking with (Eddy Coils: Stem 1set = 2 No, Bell 5 set = 10 No)	No.	1	EFD
2.29	Dot matrix marking of component (Please note :- No Orientation required at Marking Station for O-race component)	No.	1	EFD
2.30	Pick and place walking beam for transporting the Component for ECT and marking	No.	1	EFD
2.31	Anti-rust spray unit with, spraying nozzles, chamber and collection tank, Air blowing with Walking beam to transport the component.	NO.	1	EFD
3.00	Machine Accessories			

3.01	Pneumatic elements and solenoids	Set	1	FESTO
3.02	Auto lubrication, circuit and pump, collection trays	No.	1	Cenlub
3.03	Vented hood to connect to Mist collector	No.	1	EFD
3.04	Mist collector with mounting arrangement one for IH	No.	1	Span
3.05	Machine painting, labeling	Set	1	As per WIA St
3.06	SS Compact paper band filter	Set	1	SPAN
3.07	Air & Quench jet washing guns	Set	1	EFD
3.08	3-tier lamps, machine indications and controls, machine lamp	Set	1	Various
4.00	Power source and accessories			
	Power Source(type, model, frequency, outputs)			
4.01	SINAC 200/320 kW SM double output, 20-30 kHz for Bell, 9-13 kHz for male stem,	No.	1	EFD
5.00	Cooling and Quench systems & accessories			
5.01	Cooling System	Set	1	(RC 350-1000-250)/Suitable
5.02	Quench re-circulating system (Individual pumps & heat exchangers for stem and bell) with oil skimmer	Set	1	(RQ 350x2-2000-300x2)/Suitable
5.03	Magnetic separator	Set	1	Suitable
5.04	Booster pump for cooling water circuit (High Pressure)	Set	1	Grundfoss
5.05	Intermediate tank + Collection tank after filtration with circuit	Set	1	EFD
5.06	Frame for cooling systems with integral ladder – guarding	Set	1	EFD
6.00	Integration			
6.01	Integrated piping & instruments	Set	1	SS
6.02	Electrical Integration & Cabling	Set	1	EFD
7.00	Process monitoring system			
7.01	Energy Monitoring System	No.	2	EFD
7.02	Quench Flow Monitoring System	No.	4	IFM
7.03	Quench Temperature Monitoring System	No.	1	EFD
8.00	Process tooling for FOR (5 Varieties of FOR)			
8.01	Stem coil	No.	3+3	EFD
8.02	Bell Coil	No.	5+5	EFD
8.03	Fixture on feed table	No.	15+1 5	EFD

8.04	Coil setting mandrels	No.	8	EFD
8.05	Fixture at Height Checking Station	No.	5+5	EFD
8.06	Spare top center for Stem station	No.	1	EFD
9.00	Process Prove Out and Consistency Test			
9.01	Component varieties for tests			2 variety at EFD for Pre acceptance & 5 variety prove out at M/s Hyundai WIA India plant
9.02	Productivity & Consistency tests, number of components			150 No's.each Variety
9.03	Laboratory inspection			First, Middle & Last
9.04	Customer training			3 days
9.05	Machine inspection			3 days
10.00	Spare Part As per Below List			EFD
11.00	Packing			EFD

Note:

6. For cooling re-circulating system, you have to provide cooling water from your cooling tower to the heat exchangers at a temperature range of 28-32°C at a pressure of 4 to 6 bar.
7. Customer to unload the machine and position as per the layout provided by EFD at its site
8. Provide all the utilities as per the requirements of the machine.
9. Customer to provide component cutting facility during trials at customer location.
10. Important note on Price Breakup: - a) Since this is a project execution, the price break-up provided above is just notational as it is not possible to break-up the cost of engineering, training costs, costs towards warranty etc.
b) The Price breakup is just for your understanding and we request you not to consider the above prices as benchmark for any future order placed on us.

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₹ 4,55,20,000/-

(Rupees Four Crore Fifty Five Lakhs Twnty Thousand Only)

Installation & Commissioning charges extra:- ₹ 1,80,000/-

(Rupees Three Lakhs Sixty Thousand Only)

EFD Induction Private Limited

Spare Parts List as Follows:

SL.No	Item No.	Description	Qty
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3	IEN00341	ELECTRONIC CARD – SMPS	2
4	IEN00338	ELECTRONIC CARD – POWS	1
5	SIFC0021	ELECTRONIC CARD – ASMO	1
6	SIFC0019	ELECTRONIC CARD – POCC	1
7	IEN00338	ELECTRONIC CARD – POFR	2
8	SIFC0020	ELECTRONIC CARD – MIFC	1
9	SIFC0016	ELECTRONIC CARD – VCPM	1
10	SIFC0017	ELECTRONIC CARD – MGDC	2
11	SIFC0012	ELECTRONIC CARD – MGDA	1
12	IEN00698	VARISTOR	1
13	FIHH0016	HAND HELD TRANSFORMER 1200	1
14	PEH00026	TRANSFORMER THREE PHASE	1
15	SICM0030	TRANSISTOR CONTROL CABLE	4
16	IOT00680	COOLING FAN	1
17	IOT01384	ISOLATION AMPLIFIER	1
18	IOT00693	DISPLAY FOR ISOLATION AMPLIFIER	1

MACHINE FEATURES:

5. POWER SOURCE MODEL (SINAC SM)

The power source offered (SINAC 200/320 SM) is of transistorized (IGBT) design with peak output power of 320 kW intermittent duty and 200 kW in continuous duty. The power source is a double output version. The output has an operating frequency band is 20-30 kHz for Bell, 9-13 kHz for Stem hardening.

Item	Qty	Description
1		Power Source
1.1	1	Transistorised Frequency Converter (series compensated) Type SINAC 200 / 320 SM, single output Max. Output power (intermittent duty) 320 kW Output power (continuous) 200 kW Frequency range 20-30 (Bell) & 9-13 kHz (Stem) Features and Modules • Compact cabinet for indoor operation (IP 54 protection) • Main input terminal • Diode rectifier • Converter with IGBT-transistors • Output capacitor module • Electronic control device • Indirect cooling water system • External or internal power pre-selection/ regulation • Operating panel as plug - in unit with indicating instruments and transparent protection panel • LED single fault display (alarm card)
1.2	2	Water Cooled Hand Held Output Transformer.
1.3	2	Set of inductor connecting brackets, quick clamping type (manual) with groove guide for precise inductor clamping

6. RE-CIRCULATING TYPE COOLING SYSTEMS FOR QUENCH MEDIUM AND COOLING WATER

- 2.1 1 **Cooling Unit for Power Source, Output Transformer, Busbar and Coils**
(heat exchanger principle)
Type: (RC 350-1000-250)/Suitable

Water tank in stainless steel, strainer, circulating pump with motor, plate type heat exchanger, pressure gauges, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

Cooling capacity	350 MJ
Tank capacity	1000 L
Pump flow rate	250 L/min.
Pump pressure	7,5 bar

- 2.2 1 **Cooling Unit for Quench Medium**
(Heat exchanger principle)
Type: (RQ 350x2-2000-300x2)/7-5 or suitable

Water tank in stainless steel, Compact paper band filter, with strainers, circulating pump with motor, Pressure gauges, plate type heat exchanger, pressure relief valve, mains water regulation valve, completely pre-assembled on a common frame

Technical Data

Cooling capacity	350 MJ X 2
Tank capacity	2000 L
Pump flow rate for	300 L/min X2
Pump pressure	7,5 bar

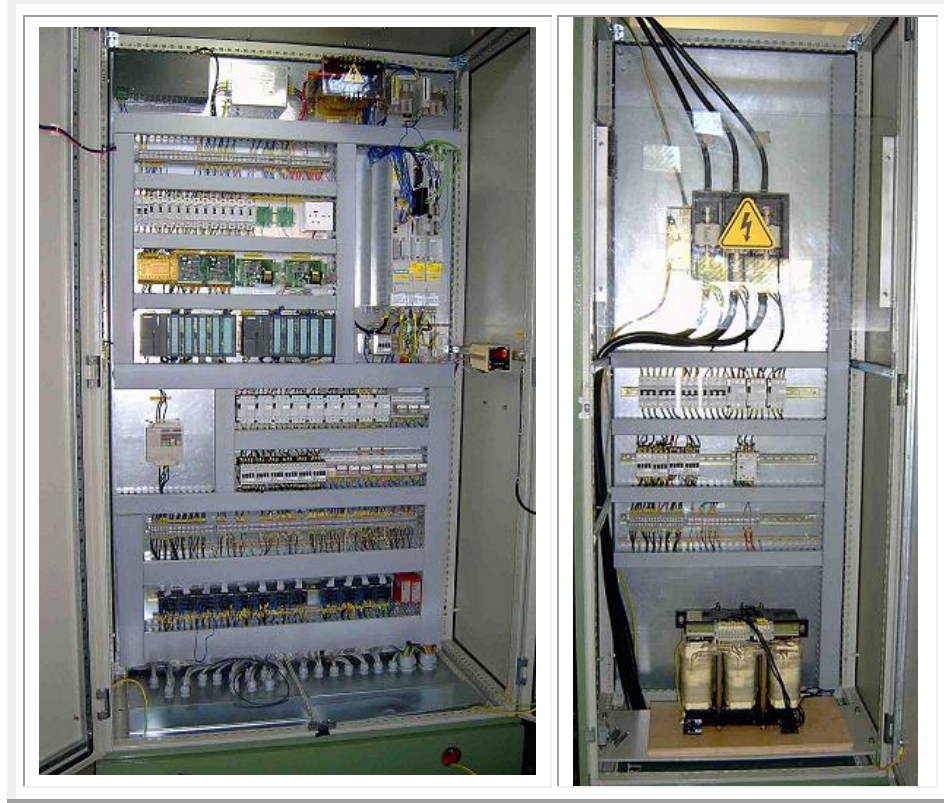
Please take notice:

The re-circulating systems must be provided with cooling water, max. 32°C, min. 4 to 6 bar pressure, flow rate equivalent to the stated circulating pumps.

Accessories

- 2.3 1 Platform to assemble cooling units on top of each other with integral ladder

7. CONTROL PANEL HOUSING CNC DRIVES, ELECTRICAL SWITCH GEAR AND DISTRIBUTION PANEL HOUSING MAINS AND FUSES



8. PNEUMATIC CONTROL PANEL





COMMERCIAL TERMS:

We request you to include the below statements in you PO.

“ Termination for convenience (cancellation) of a Purchase Order

In the case of a Purchase Order for the delivery of customized equipment and subject to the conditions below, EFD Induction allows for termination for convenience by the Purchaser.

In such case, EFD Induction shall be entitled to invoice to the Purchaser the work carried out and materials or services received to the date for the execution of the purchase order as well as the work, materials or services already ordered with suppliers to the extent that these cannot be cancelled. In addition, EFD Induction shall be entitled to a compensation for profit equal to 15% of the balance between the aforesaid amount and the Purchase Price.

To avoid the determination of the degree of advancement or the works contracted and to limit the Purchaser's risk, EFD Induction prefers to enter into a cancellation scheme to apply in the case of termination for convenience. Such scheme shall be agreed in advance and become integral part of the Purchase Order.

Upon payment of the aforesaid amounts, any work carried out or materials received for the Purchaser Order shall become property of the Purchaser. EFD Induction shall act in good faith as regards the buy-back for re-use or re-sale of work or materials that the Purchaser cannot use.

Regarding the delivery of standard products in the context of purchase agreements or work performance contracts, termination for convenience by the Purchaser remains at the discretion of EFD Induction. When a termination is accepted, EFD Induction shall be entitled to compensation as stated above.”

1.0 Price Terms:

- ☐ Ex-works without packing and forwarding
- ☒ Ex-works with packing and forwarding
- ☐ Free delivery up to customer's site
- ☐ Free delivery to transport office
- ☐ Free delivery to nearest port

All taxes and duties will be extra at actual and at rates prevailing at the time of despatch.

2.0 Packing Type:

- ☒ Wooden platform and machine covered with bubble sheet
- ☐ Carton box packing with nylon straps
- ☐ Complete Wooden Box Packing for transport by road / rail
- ☐ Sea-worthy packing

3.0 Installation & Commissioning Charges:

- ☒ Included ☐ Not Included

4.0 Scope of other charges:

- | | | |
|------------------------|--|--|
| Transportation Charges | ☐ Included ☐ Not | ☒ Customer's Scope |
| Transit Insurance | ☐ Included ☐ Not | ☒ Customer's Scope |
| Others | ☐ Included ☐ Not | ☒ Customer's Scope |

5.0 Delivery Period: 9.5 Months

The delivery period is counted from the date of our Order Acceptance. The Order Acceptance will be sent within Five days of EFD receiving the printed copy of the Purchase Order/LOI. The order acceptance is to confirm preliminary acceptance of the order. However, if there are any deviations found in order after detailed contract review it will be addressed in a separate correspondence from EFD. If due to such changes, the delivery date has to be extended it shall be done as per EFD's request.

6.0 Penalty Clause: - LD charges 0.5% per week & up to maximum 5% of basic value. Penalty

Applicable after two weeks of Delivery date.

Note:

LD will not be applicable for duration which is attributed to the activity to be carried out by M/s. Hyundai WIA, some of the crucial / Major activities are listed below:-

- i) Delay in DAP approval.
- ii) Delay in advance Payment.
- iii) Delay in providing samples for trials.
- iv) Delay in visiting for Inspection and dispatch clearance.
- V) Delay in providing correct drawing of component to design tooling.

7.0 Payment Terms:

30% Advance payment against corporate guarantee.
EFD Induction Private Limited



50% With GST against delivery at M/s Hyundai Wia, India.

20% Against Installation & Commissioning at M/s Hyundai Wia, India.

8.0 VALIDITY: Thirty days from the date of this offer.

9.0 PRE - ACCEPTANCE TEST:

Prices include the demonstration of process for maximum of two variety of component. Work pieces must be identical to the parts which will be used for final production at buyer's works. Test pieces to be demonstrated must be available in sufficient quantity at our works at least 12 weeks before the agreed delivery date. If receipt of test pieces is delayed, scheduled delivery may take place without demonstration or written confirmation of the buyer. The same clause regarding delivery will apply if clearance is delayed even after successful process demonstration.

10.0 FINAL ACCEPTANCE:

Price includes the demonstration of the process for maximum of 5 variety of FOR component. Additional prove-outs will be undertaken at extra charges.

11.0 WARRANTY:

12 months from the date of Commissioning or 18 Months from the date of Dispatch whichever comes first. Warranty for CNC 18 Months from the date of Commissioning or 24 Months from the date of dispatch whichever comes first. The warranty does not cover consumables, toolings and other safety elements.

Usage of any spurious parts in EFD machine by the customer, then immediately warranty will be void and EFD is not responsible for any breakdowns.

12.0 RETENTION OF TITLE:

Title to all goods will be retained with us till our each and every claim against the buyer originating in the course of the contract has been fully satisfied.

13.0 COPY RIGHT:

The specification and technology contained in our proposal is the property of EFD. Passing on this information to third parties as well as unauthorised utilisation by the buyer himself are both not permitted.

14.0 SUBSEQUENT ALTERATIONS TO ORDER:

EFD Induction Private Limited



If the order is altered, the price and delivery time may be revised.

15.0 DISPUTES:

In accordance with UN 188 A with addendum of 1987.

16.0 CANCELLATION:

Actual project costs up to the date of cancellation is to be paid by the customer, plus 18% of total contract value as a compensation for production readjustment costs and loss of project income

***“Any deviations to the above terms are valid only
if EFD has accepted in writing prior to order release or during order acceptance.”***